

Construction Safety NLP — Executive Report

Author: Yenlik Gaisina | Portfolio NLP case study | May 2026

Problem Statement

Construction is one of the highest-risk industries worldwide. Each year thousands of OSHA accident narratives are filed, but they remain mostly unread because reviewing each one manually is slow and costly. The risk: recurring failure patterns stay invisible to safety managers until another serious incident occurs.

Dataset

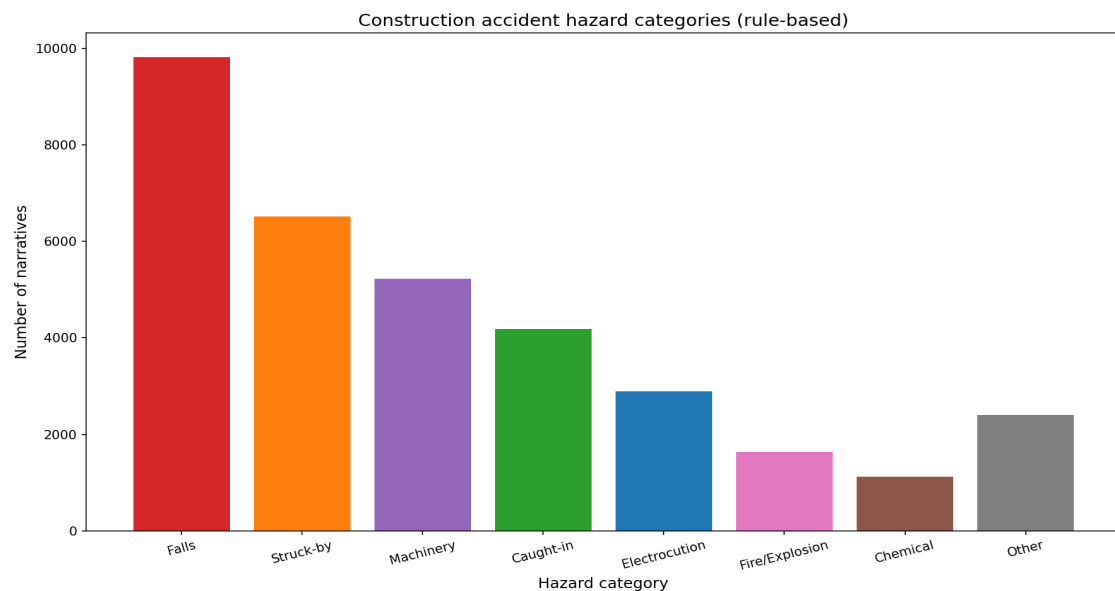
OSHA Accident and Injury Data 2015–2017 (Kaggle). Approximately 25,000 free-text incident narratives. Raw data is not redistributed; users download directly from Kaggle.

Methods

A standard NLP pipeline was applied: cleaning, lowercase, stopword removal, lemmatisation, word-frequency analysis, word cloud, rule-based hazard classification (OSHA Fatal Four + extensions), BERTopic for semantic topic discovery, and an LDA benchmark for confirmation. Visualisations and recommendations were derived from the outputs.

Key Findings

1. **Falls dominate** — around one in three narratives mention falls from scaffolds, ladders, or roofs.
2. **Struck-by and caught-in** events form the second tier, frequently involving machinery, falling objects, or trench walls.
3. **BERTopic surfaces fine-grained themes** that simple keyword grouping misses, including electrical contact, trench collapse, and forklift tip-overs.
4. **LDA confirms** the BERTopic themes, increasing confidence in the patterns.



Business Value

NLP converts unread incident reports into a quantified, prioritised safety dashboard. Safety teams can move from reactive incident review to proactive, theme-led intervention — saving review hours and reducing repeat incidents.

Safety Recommendations

1. **Fall protection audits** on scaffolds, ladders and roof edges every quarter.
2. **Mandatory struck-by training** for ground crews near heavy equipment.
3. **Trench shoring sign-off** for any excavation deeper than 5 ft.
4. **Lock-out / tag-out enforcement** before energised electrical work.
5. **Toolbox talks targeted to BERTopic themes** so briefings reflect actual failure patterns.

Limitations

The dataset is biased toward serious/reportable incidents and may under-represent near-misses. Narratives are written by investigators, so phrasing is consistent but may bury root causes. Topic models surface correlations, not causation. The 2015–2017 window pre-dates several recent regulatory changes.

Conclusion

NLP turns a stack of unread incident reports into a quantified, prioritised safety dashboard. By combining rule-based hazard tagging with BERTopic and LDA, recurring failure patterns can be identified at scale and translated into actionable interventions.